# P2 Chapter 10: Numerical Methods

## Newton-Raphson

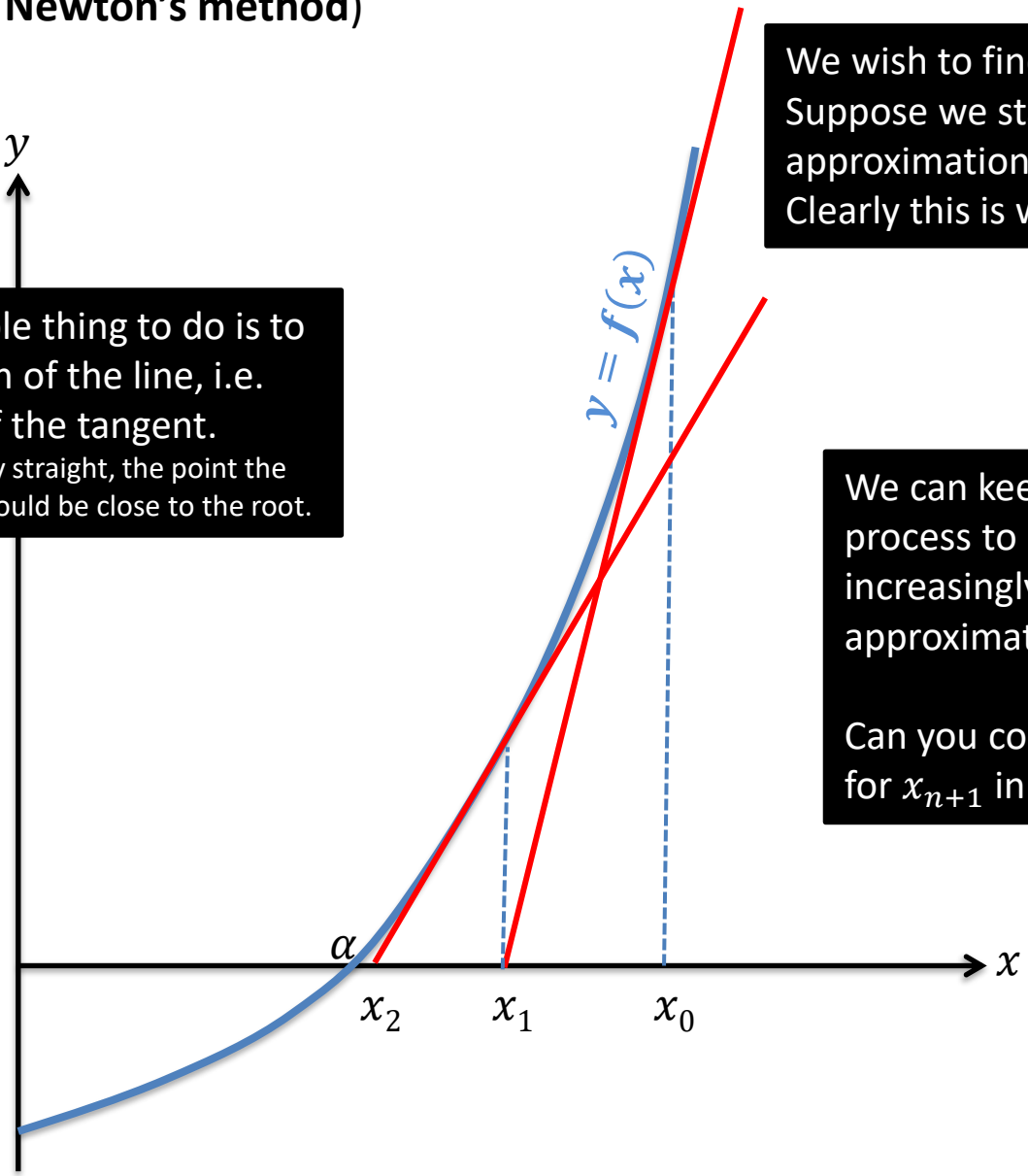
# The Newton-Raphson Process

(also known as **Newton's method**)

We wish to find the root $\alpha$. Suppose we start with the indicated approximation of the root, $x_0$. Clearly this is well off the mark!

A seemingly sensible thing to do is to follow the direction of the line, i.e. use the gradient of the tangent.

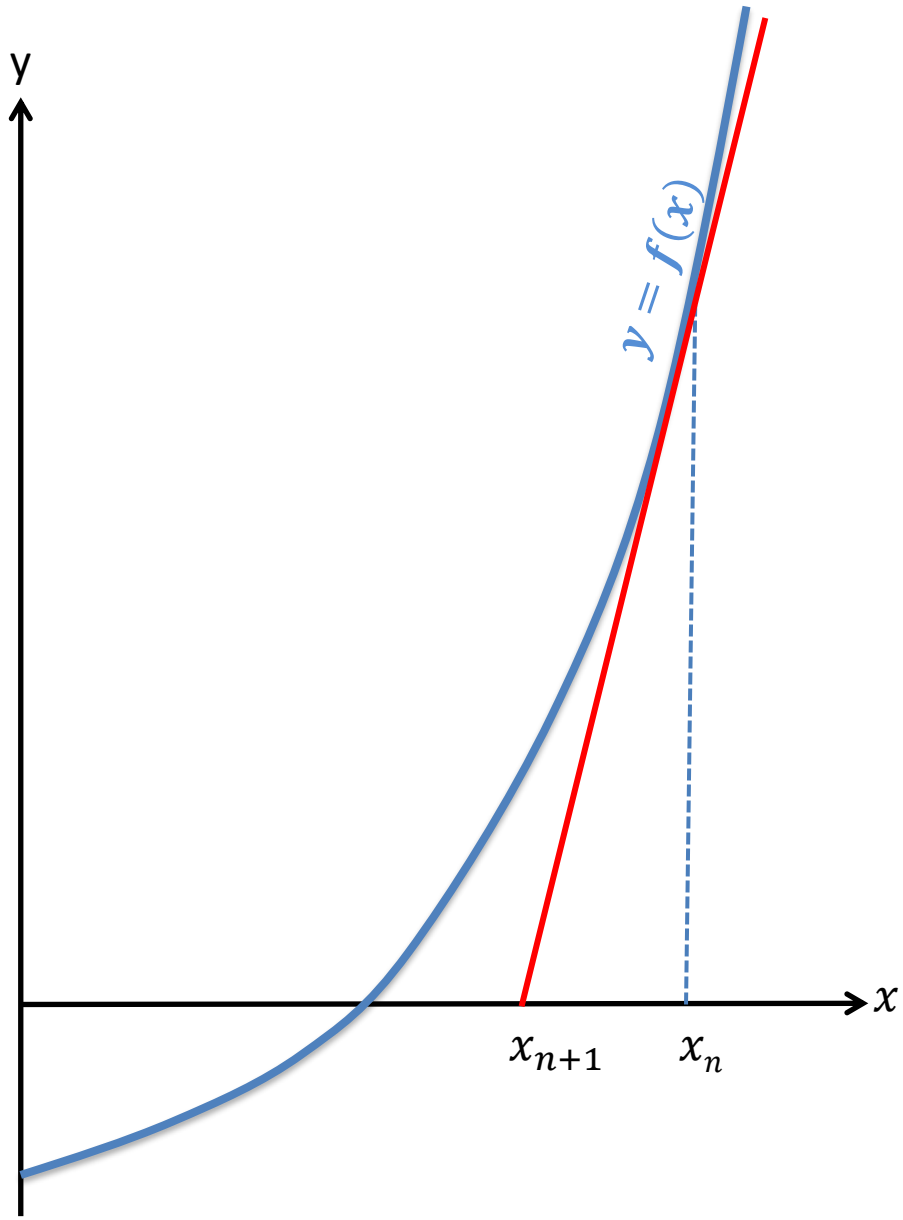
If the line was reasonably straight, the point the tangent hits the $x$-axis would be close to the root.




We can keep repeating this process to (hopefully) get increasingly accurate approximations.

Can you come up with a formula for $x_{n+1}$ in terms of $x_n$?

# The Newton-Raphson Process




**Formula:**

Using Year 1 coordinate geometry:

$$\boldsymbol{y - f(x_n) = f'(x_n)(x - x_n)}$$

But we're interested when
$x = x_{n+1}$ and $y = 0$

$$\boldsymbol{-f(x_n) = f'(x_n)(x_{n+1} - x_n)}$$

which rearranges to give:

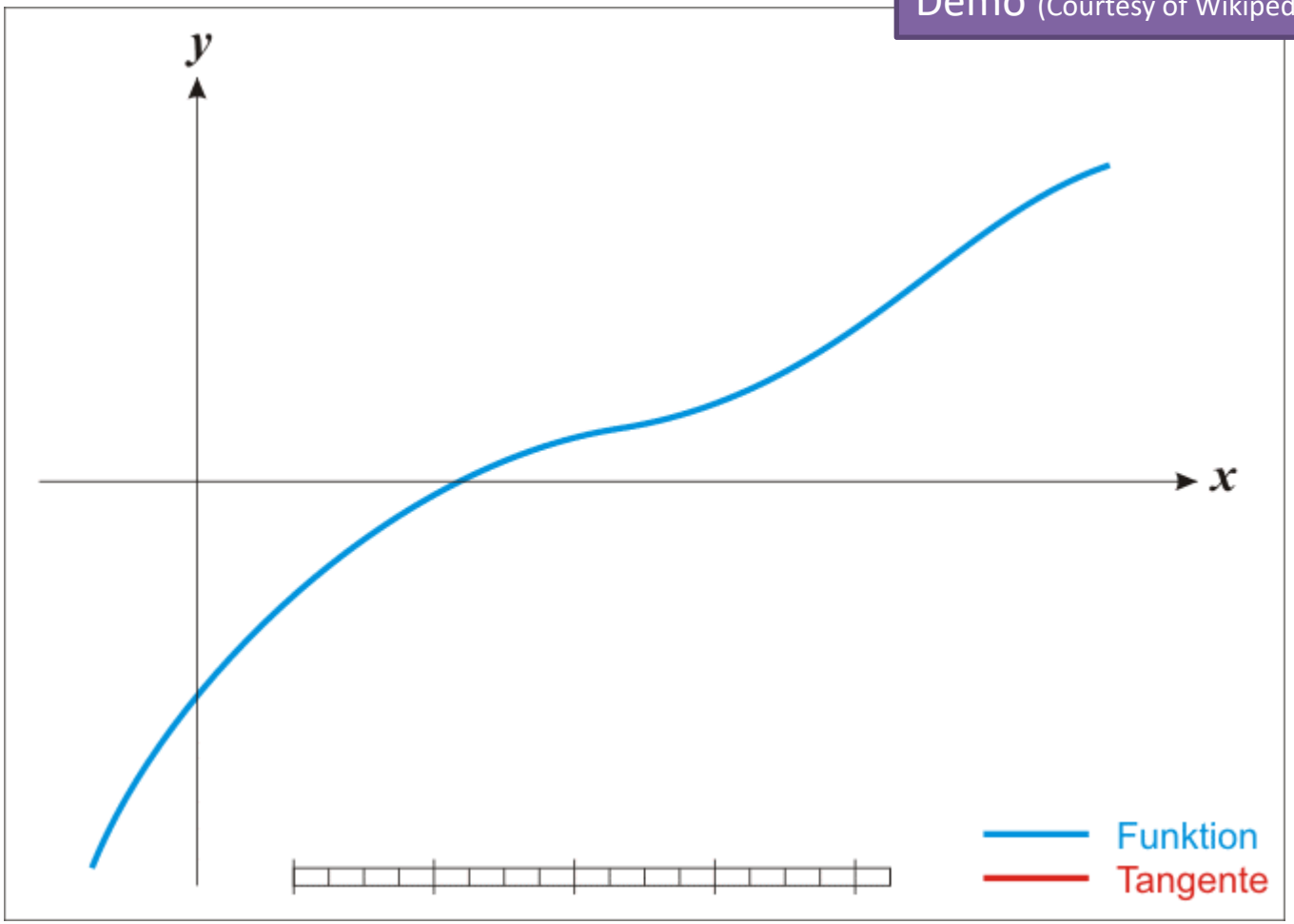
**Newton-Raphson Process:**

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

**Exam Note**: Pearson didn't include the above proof in their own textbook, so I can only presume you won't be tested on the proof.

# A nice animation…

Demo (Courtesy of Wikipedia)




This is German, but I'll presume you're not an idiot.

# Example

Returning to our original example: $\boldsymbol{x} = \mathbf{cos}(\boldsymbol{x})$, say letting $x_0 = 0.5$

*(Note: Recall that differentiation assumes radians)*

?

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# Example

Returning to our original example: $\boldsymbol{x = \cos(x)}$, say letting $x_0 = 0.5$

*(Note: Recall that differentiation assumes radians)*

**Let** $\boldsymbol{f(x) = x - cos(x)}$ ← This ensures we have an equation of the form $f(x) = 0$

$$\boldsymbol{f'(x) = 1 + sin\, x}$$

$$\boldsymbol{\therefore x_{n+1} = x_n - \frac{x_n - cos(x_n)}{1 + sin(x_n)}}$$

$$\boldsymbol{x_1 = 0.5 - \frac{0.5 - cos(0.5)}{1 + sin(0.5)} = 0.7552224171}$$

$$\boldsymbol{x_2 = 0.7391412}$$

$$\boldsymbol{x_3 = 0.7390851}$$

After merely three iterations, our approximation is accurate to 7 decimal places. Holy smokes Batman!

**Tip**: To perform iterations quickly, do the following on your calculator:

[0.5] [=]
[ANS] – (ANS – cos(ANS))/(1 + sin(ANS))
Then spam [=].

# Quickfire Questions

Using the Newton-Raphson process, state the recurrence relation for the following functions:

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| | |
|---|---|
| $f(x) = x^3 - 2$ | ? |
| $f(x) = \tan x$ | ? |
| $f(x) = x^2 - x - 1$ | ? |

# Quickfire Questions

Using the Newton-Raphson process, state the recurrence relation for the following functions:

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$f(x) = x^3 - 2$ ⇒ $\boldsymbol{x_{n+1} = x_n - \frac{(x_n)^3 - 2}{3(x_n)^2}}$

$f(x) = \tan x$ ⇒ $\boldsymbol{x_{n+1} = x_n - \frac{\tan x_n}{\sec^2 x_n} = x_n - \frac{1}{2}\sin(2x_n)}$

$f(x) = x^2 - x - 1$ ⇒ $\boldsymbol{x_{n+1} = x_n - \frac{(x_n)^2 - x_n - 1}{2x_n - 1}}$

# Example Exam Question

Edexcel FP1 June 2013(R) Q3c

$$f(x) = \frac{1}{2}x^4 - x^3 + x - 3$$

...

The equation $f(x) = 0$ has a root $\beta$ in the interval $[-2, -1]$.

(c) Taking $-1.5$ as a first approximation to $\beta$, apply the Newton-Raphson process once to $f(x)$ to obtain a second approximation to $\beta$.
Give your answer to 2 decimal places.

**(5)**

?

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

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$$\boxed{x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}}$$

$$\boldsymbol{x_{n+1} = x_n - \frac{\frac{1}{2}(x_n)^4 - (x_n)^3 + x_n - 3}{2(x_n)^3 - 3(x_n)^2 + 1}}$$

$$\boldsymbol{\beta_1 = -1.5 - \frac{1.40625}{-12.5} = -1.3875}$$

# Test Your Understanding

Edexcel FP1 Jan 2010 Q2c

$$f(x) = 3x^2 - \frac{11}{x^2}.$$

(*c*) Taking 1.4 as a first approximation to $\alpha$, apply the Newton-Raphson procedure once to f($x$) to obtain a second approximation to $\alpha$, giving your answer to 3 decimal places.

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# Test Your Understanding

Edexcel FP1 Jan 2010 Q2c

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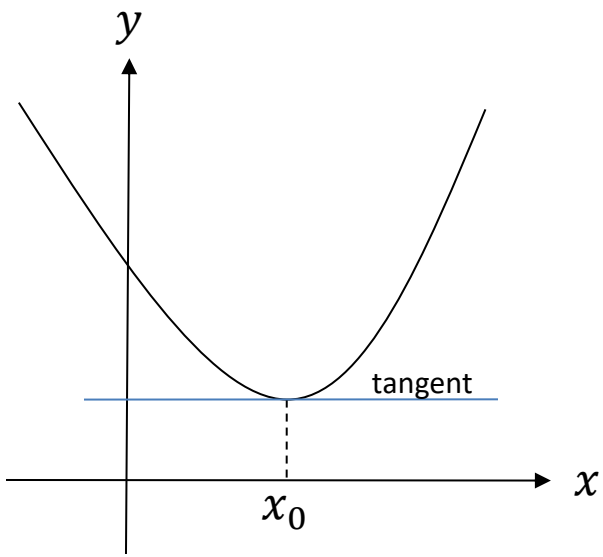
**(5)**

| | |
|---|---|
| (c) $f'(x) = 6x + 22x^{-3}$ | M1 A1 |
| $x_1 = x_0 - \frac{f(x_0)}{f'(x_0)} = 1.4 - \frac{0.268}{16.417}, \quad = 1.384$ | M1 A1, A1 |
| | (5) |

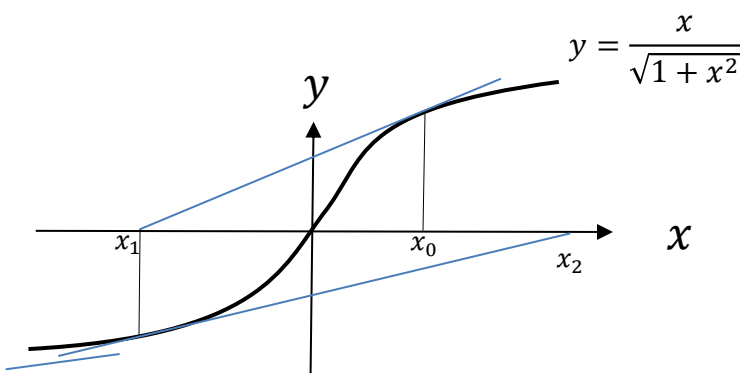
# When does Newton-Raphson fail?




$$\boxed{x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}}$$

**If the starting value $x_0$ was the stationary point**, then $f'(x_0) = 0$, resulting in a division by 0 in the above formula.

Graphically, it is because the tangent will never reach the $x$-axis.




Newton-Raphson also suffers from the same drawbacks as solving by iteration, in that it's possible for the values of $x_i$ to **diverge**.
In this example, the $x_i$ oscillate either side of 0, but gradually getting further away from $\alpha = 0$.

# Exercise 10.3

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# Homework Exercise

**1** $f(x) = x^3 - 2x - 1$

**a** Show that the equation $f(x) = 0$ has a root, $\alpha$, in the interval $1 < \alpha < 2$.

**b** Using $x_0 = 1.5$ as a first approximation to $\alpha$, apply the Newton–Raphson procedure once to $f(x)$ to find a second approximation to $\alpha$, giving your answer to 3 decimal places.

**2** $f(x) = x^2 - \frac{4}{x} + 6x - 10, x \neq 0.$

**a** Use differentiation to find $f'(x)$. **(2 marks)**

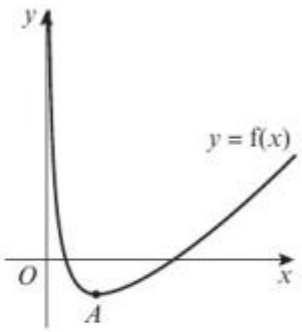
The root, $\alpha$, of the equation $f(x) = 0$ lies in the interval $[-0.4, -0.3]$.

**b** Taking $-0.4$ as a first approximation to $\alpha$, apply the Newton–Raphson process once to $f(x)$ to obtain a second approximation to $\alpha$. Give your answer to 3 decimal places. **(4 marks)**

**3** The diagram shows part of the curve with equation

$y = f(x)$, where $f(x) = x^{\frac{3}{2}} - e^{-x} + \frac{1}{\sqrt{x}} - 2, x > 0.$

The point $A$, with $x$-coordinate $q$, is a stationary point on the curve.
The equation $f(x) = 0$ has a root $\alpha$ in the interval $[1.2, 1.3]$.




**a** Explain why $x_0 = q$ is not suitable to use as a first approximation when applying the Newton–Raphson method. **(1 mark)**

**b** Taking $x_0 = 1.2$ as a first approximation to $\alpha$, apply the Newton–Raphson process once to $f(x)$ to obtain a second approximation to $\alpha$. Give your answer to 3 decimal places. **(4 marks)**

**4** $f(x) = 1 - x - \cos(x^2)$

**a** Show that the equation $f(x) = 0$ has a root $\alpha$ in the interval $1.4 < \alpha < 1.5$. **(1 mark)**

**b** Using $x_0 = 1.4$ as a first approximation to $\alpha$, apply the Newton–Raphson procedure once to $f(x)$ to find a second approximation to $\alpha$, giving your answer to 3 decimal places. **(4 marks)**

**c** By considering the change of sign of $f(x)$ over an appropriate interval, show that your answer to part **b** is correct to 3 decimal places. **(2 marks)**

# Homework Exercise

**5** $f(x) = x^2 - \frac{3}{x^2}, x \geqslant 0$

**a** Show that a root $\alpha$ of the equation $f(x) = 0$ lies in the interval [1.3, 1.4]. **(1 mark)**

**b** Differentiate $f(x)$ to find $f'(x)$. **(2 marks)**

**c** By taking 1.3 as a first approximation to $\alpha$, apply the Newton–Raphson process once to $f(x)$ to obtain a second approximation to $\alpha$. Give your answer to 3 decimal places. **(3 marks)**

**6** $y = f(x)$, where $f(x) = x^2 \sin x - 2x + 1$. The points $P$, $Q$, and $R$ are roots of the equation. The points $A$ and $B$ are stationary points, with $x$-coordinates $a$ and $b$ respectively.

**a** Show that the curve has a root in each of the following intervals:

**i** [0.6, 0.7] **(1 mark)**

**ii** [1.2, 1.3] **(1 mark)**

**iii** [2.4, 2.5] **(1 mark)**

**b** Explain why $x_0 = a$ is not suitable to use as a first approximation to $\alpha$ when applying the Newton–Raphson method to $f(x)$. **(1 mark)**

**c** Using $x_0 = 2.4$ as a first approximation, apply the Newton–Raphson method to $f(x)$ to obtain a second approximation. Give your answer to 3 decimal places. **(4 marks)**

**7** $f(x) = \ln(3x - 4) - x^2 + 10, x > \frac{4}{3}$

**a** Show that $f(x) = 0$ has a root $\alpha$ in the interval [3.4, 3.5]. **(2 marks)**

**b** Find $f'(x)$. **(2 marks)**

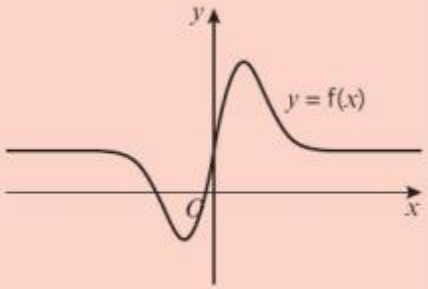
**c** Taking 3.4 as a first approximation to $\alpha$, apply the Newton–Raphson procedure once to $f(x)$ to obtain a second approximation for $\alpha$, giving your answer to 3 decimal places. **(3 marks)**

# Homework Exercise

**Challenge**

$f(x) = \frac{1}{5} + xe^{-x^2}$

The diagram shows a sketch of the curve $y = f(x)$. The curve has a horizontal asymptote at $y = \frac{1}{5}$.




**a** Prove that the Newton–Raphson method will fail to converge on a root of $f(x) = 0$ for all values of $x_0 > \frac{1}{\sqrt{2}}$

**b** Taking $-0.5$ as a first approximation, use the Newton–Raphson method to find the root of $f(x) = 0$ that lies in the interval $[-1, 0]$, giving your answer to 3 d.p.

# Homework Answers

**1** **a** $f(1) = -2$, $f(2) = 3$. There is a sign change in the interval $1 < \alpha < 2$, so there is a root in this interval.
**b** $x_1 = 1.632$

**2** **a** $f'(x) = 2x + \frac{4}{x^2} + 6$ **b** $-0.326$

**3** **a** It's a turning point, so $f'(x) = 0$, and you cannot divide by zero in the Newton–Raphson formula.
**b** 1.247

**4** **a** $f(1.4) = -0.020\ldots$, $f(1.5) = 0.12817\ldots$ As there is a change of sign in the interval, there must be a root $\alpha$ in this interval.
**b** $x_1 = 1.413$
**c** $f(1.4125) = -0.00076\ldots$, $f(1.4135) = 0.0008112\ldots$.

**5** **a** $f(1.3) = -0.085\ldots$, $f(1.4) = 0.429\ldots$ As there is a change of sign in the interval, there must be a root $\alpha$ in this interval.
**b** $f'(x) = 2x + \frac{6}{x^3}$ **c** 1.316

**6** **a** $f(0.6) = 0.0032\ldots > 0$, $f(0.7) = -0.0843\ldots < 0$. Sign change implies root in the interval.
$f(1.2) = -0.0578\ldots < 0$, $f(1.3) = 0.0284\ldots > 0$. Sign change implies root in the interval.
$f(2.4) = 0.0906\ldots > 0$, $f(2.5) = -0.2595\ldots < 0$. Sign change implies root in the interval.
**b** It's a turning point, so $f'(x) = 0$, and you cannot divide by zero in the Newton–Raphson formula.
**c** 2.430

**7** **a** $f(3.4) = 0.2645\ldots > 0$, $f(3.5) = -0.3781\ldots. < 0$. Sign change implies root in the interval.
**b** $f'(x) = \frac{3}{3x - 4} - 2x$ **c** 3.442

**Challenge**

**a** From the graph, $f(x) > 0$ for all values of $x > 0$. Note also that $xe^{-x^2} > 0$ when $x > 0$. So the same must be true for $x > \frac{1}{\sqrt{2}}$.

$f'(x) = e^{-x^2}(1 - 2x^2) = 0 \Rightarrow x = \frac{1}{\sqrt{2}}$

So $f'(x) < 0$ for $x > \frac{1}{\sqrt{2}}$.

$x_{n+1} = x_n - \frac{f(x)}{f'(x)}$ is an increasing sequence as $f(x) > 0$ and $f'(x) < 0$, for $x > \frac{1}{\sqrt{2}}$. Therefore the Newton–Raphson method will fail to converge.

**b** $-0.209$